

Practical: 3

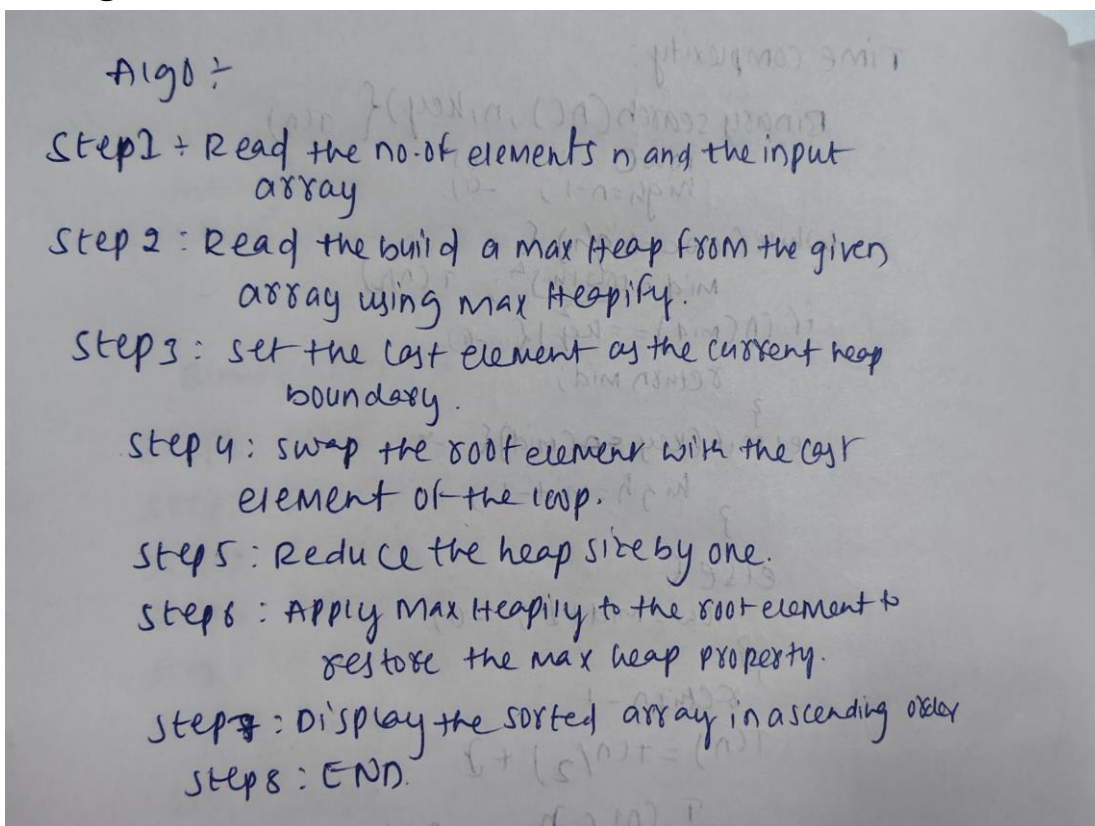
Implementation of max-heap sort algorithm.

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Aim: To Implement of max-heap sort algorithms.

MAX-HEAP SORT

Algorithm:



Program:

```
#include <iostream>
using namespace std;
void heapify(int arr[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
    if (left < n && arr[left] > arr[largest]) {
        largest = left;
    }
    if (right < n && arr[right] > arr[largest]) {
        largest = right;
    }
    if (largest != i) {
        swap(arr[i], arr[largest]);
        heapify(arr, n, largest);
    }
}
void heapSort(int arr[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--) {
        heapify(arr, n, i);
    }
    for (int i = n - 1; i > 0; i--) {
        swap(arr[0], arr[i]);
        heapify(arr, i, 0);
    }
}
int main() {
    int arr[] = {4, 10, 3, 5, 1};
    int n = 5;
    heapSort(arr, n);
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Output:

```

Output
Enter the number of elements: 5
Enter 5 elements:
3
1
6
4
2
Array after Heap Sort: 1 2 3 4 6
=== Code Execution Successful ===

```

Time Complexity:

Time complexity:

```

void maxHeapify(int arr[], int size, int i) → n
if (largest != i) → 2
swap(arr[largest], arr[i]) → 2
maxHeapify(arr, size, largest) → log n
void heapSort(int arr[], int size)
for (int i = size/2 - 1; i >= 0; i--)
maxHeapify(arr, size, i) → n
for (int i = size - 1; i > 0; i--)
swap(arr[0], arr[i]) → n
maxHeapify(arr, size, i) → log n

```

Best case:

```

for i = n/2 to 1
maxHeapify(arr, i)
for i = n to 2
swap(arr[i], arr[1])
maxHeapify(arr, i-1, 1) → log n
T(n) = O(n) + n * O(log n)
T(n) = O(n) + O(n log n)
T(n) = O(n log n)

```

Average case:

```

for i = n to 2
swap(i, 1)
maxHeapify(1, log n)
T(n) = O(n) + 1 * log n
T(n) = O(n) + O(n log n)
T(n) = O(n log n)

```

Worst case:

```

maxHeapify root element of heap
→ log n
maxHeapify(i) → log n
T(n) = n * log n
T(n) = O(n log n)

```

Conclusion:

Hence, from this practical of Max Heap Sort, the algorithm and program have been executed successfully.